Evaluation & responsible modelling

24 March 2026

LONDON SCHOOL of HYGIENE &TROPICAL MEDICINE

Welcome

10:00 “(Ir)responsible” keywords

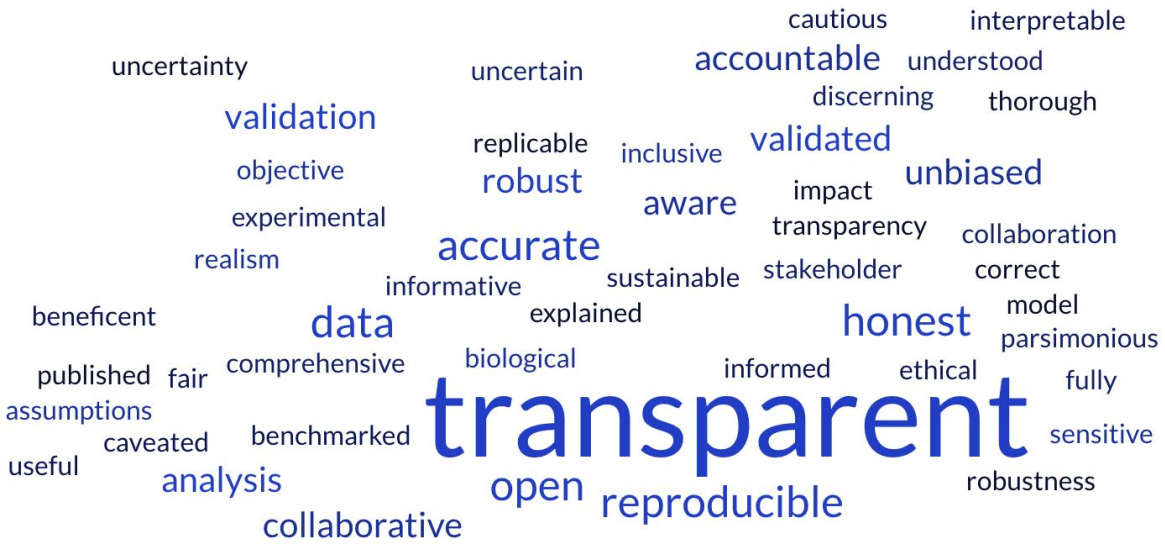
10:15 Introduction & aims

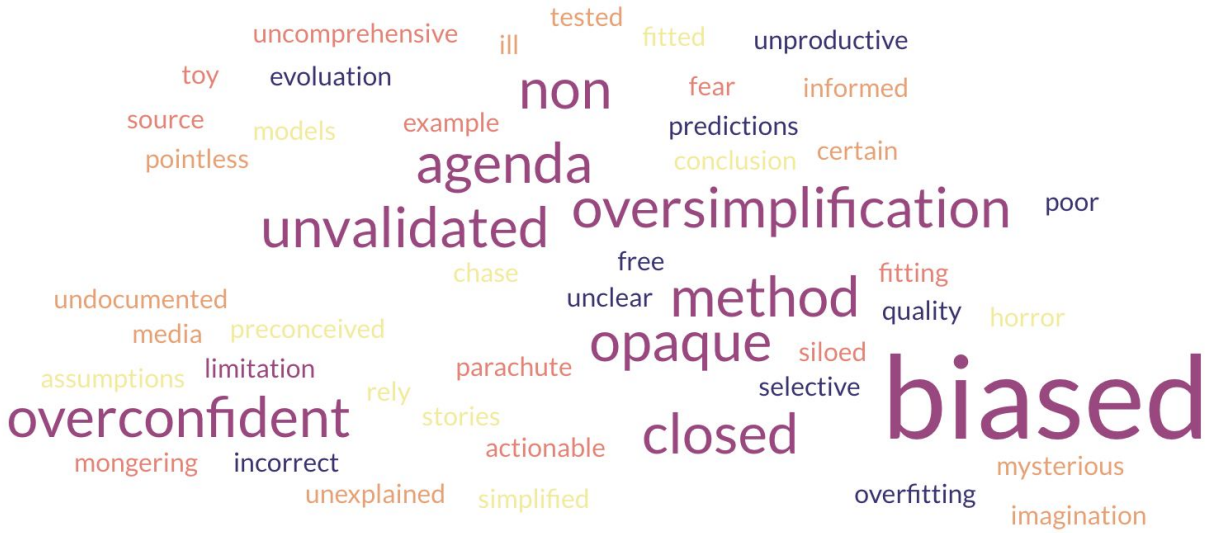
Wifi access

Network
lshtm-visitor

Username
lsh1701639@lshtm.ac.uk

Password
predegree8diplopod





Introduction

10:00 “(Ir)responsible” keywords

10:15 Introduction & aims
Katharine Sherratt, Sebastian Funk, Erica Thompson

10:30 Epidemiology & scientific responsibility

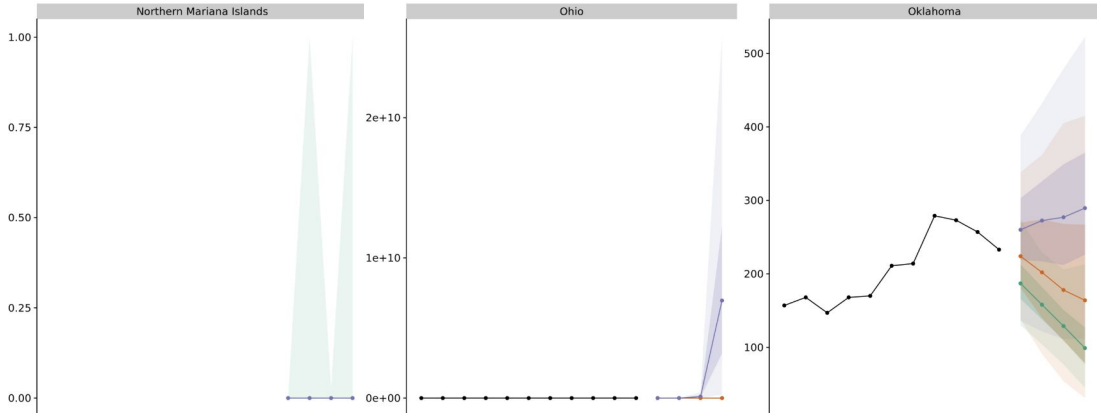
13:30 Public health & social responsibility

16:00 Implementation: priorities and next steps

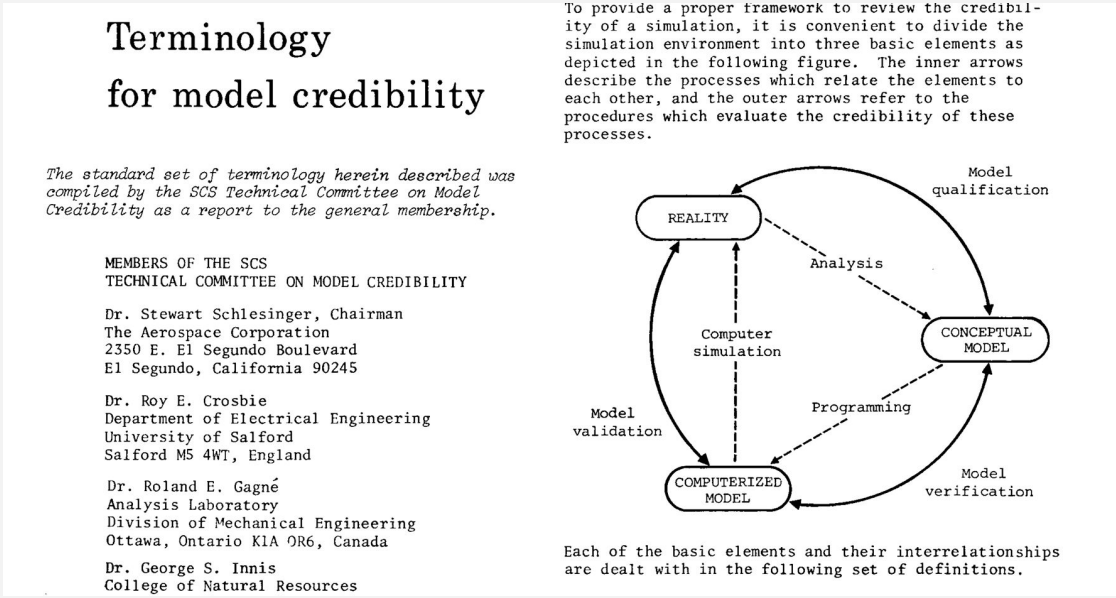
Introduction

Background: *crisis collaboration*

- Epi, dengue, COVID-19
- Multi-model collaborations (UK, US, EU)



many methods of being wrong:
COVID-19 deaths in 3 US states, Feb 2021



Terminology
for model credibility

The standard set of terminology herein described was compiled by the SCS Technical Committee on Model Credibility as a report to the general membership.

MEMBERS OF THE SCS
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To provide a proper framework to review the credibility of a simulation, it is convenient to divide the simulation environment into three basic elements as depicted in the following figure. The inner arrows describe the processes which relate the elements to each other, and the outer arrows refer to the procedures which evaluate the credibility of these processes.

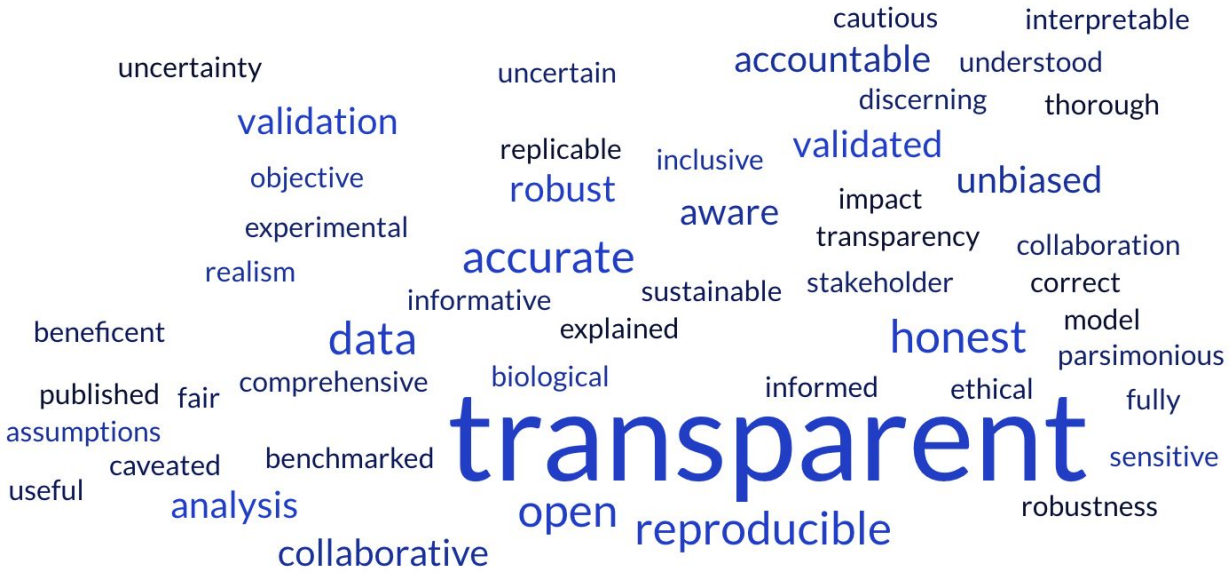
Each of the basic elements and their interrelationships are dealt with in the following set of definitions.

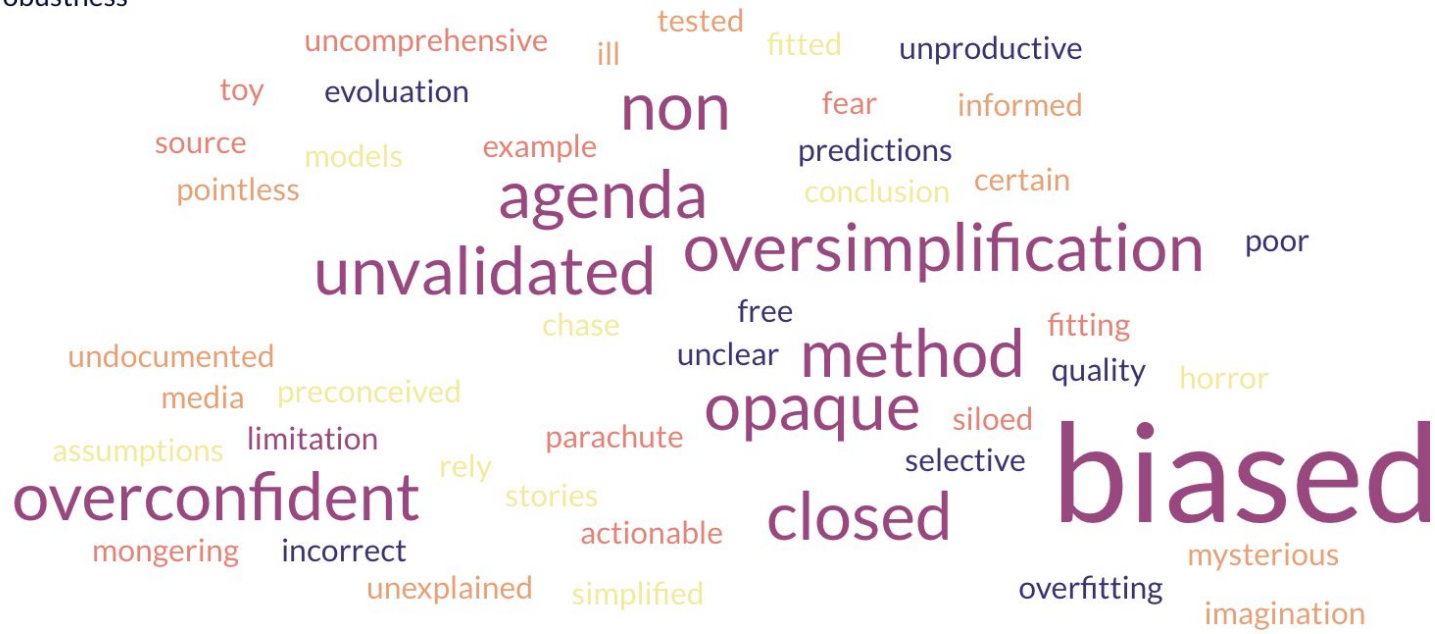
Model comparison

Evaluation

Ethics

Introduction





Agenda

10:00 “(Ir)responsible” keywords

10:15 Introduction & aims

10:30 Epidemiology & scientific responsibility

13:30 Public health & social responsibility

16:00 Implementation: priorities and next steps

Epidemiology & scientific responsibility

LONDON SCHOOL of HYGIENE &TROPICAL MEDICINE

10:30 Opening: Modelling & scientific responsibility
Sebastian Funk

10:40 Evaluation across epidemiological contexts
Panel: Matt Keeling, Kaitlyn Johnson, Jon Mellor

11:10 Implications for scientific responsibility
Nested group discussion

11:30 Coffee break

11:45 Training scientifically responsible modellers
Small group co-design

Public health & social responsibility

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Session structure

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30 mins Opening | Panel

20 mins 1 min point | 2 mins pairs| 5mins table discussion
10 minute whole group

30 mins Group work
use the discussion and share resources you know already
note what's unanswered and resources you might be missing

10 mins Review
nominate a note-taker and a speaker for your group

Introductions

5 mins

Introduce yourself to your neighbour

Your neighbour introduces you to the table

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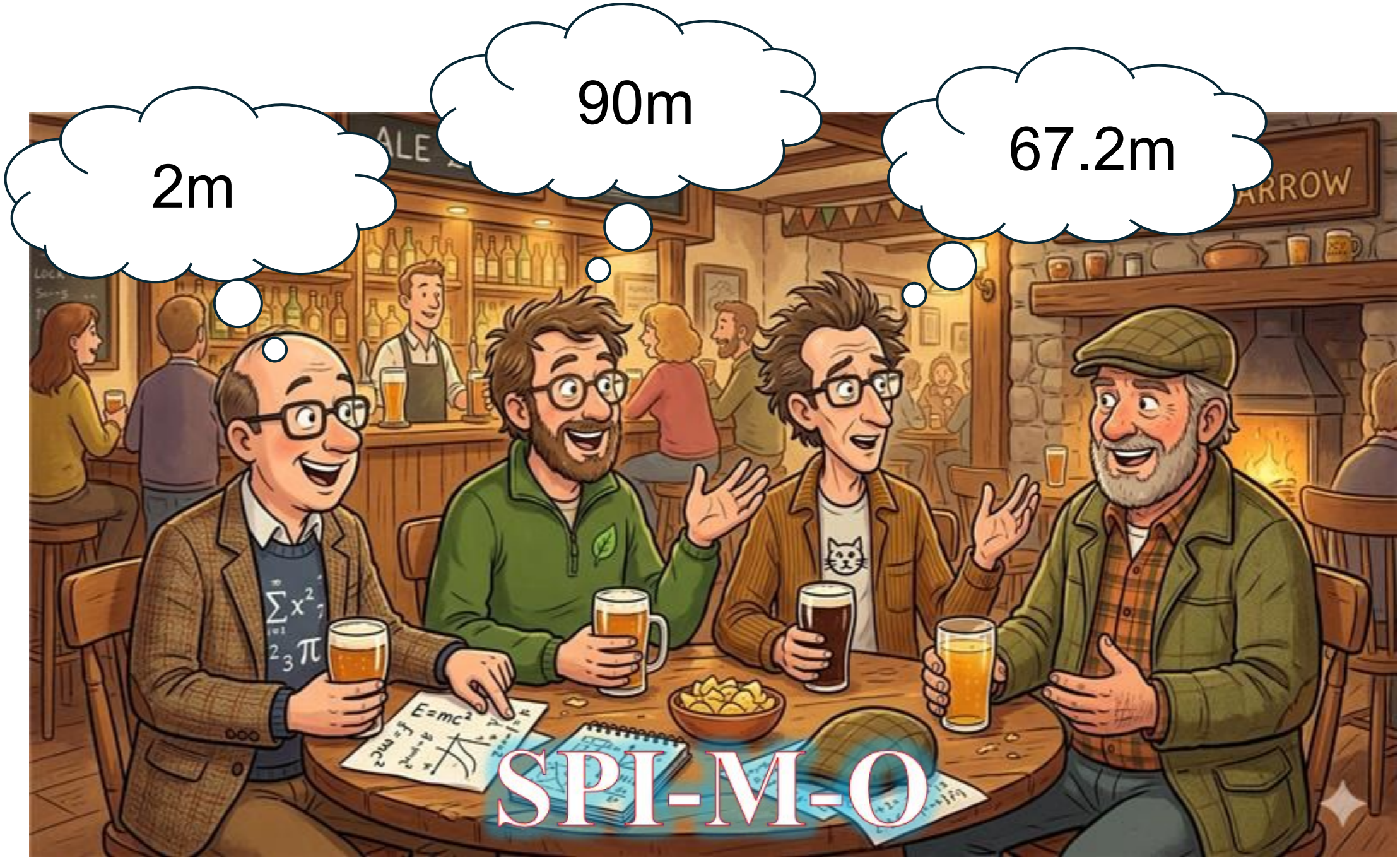
2m
90m
67.2m
SPI-M-O

Lessons from SPI-M-O

1. **It is not the modeller's role to set policy** – we provide important on an important but single aspect of the decision-making. This insulates the modeller from the consequences.
2. **Advice is a team effort**, both within institutions and on the committees. This gives shared responsibility & shared success.
3. **Be confident in your own answers**. Thinking "outside the box" is OK, it can move the science forward. But it is usually better to produce a sensible answer as well.
4. **"*Tell Me Why I'm Wrong*".** Encourage friendly criticism.
5. **Stupid questions lead to stupid answers**. Need to train policy-makers and policy-advisors to ask meaningful questions.
6. **Systematically review previous results**, and understand their value.

Lessons for New Modellers

1. Modelling is 1% inspiration (writing the equations) and 99% perspiration (getting the MCMC to converge).
2. **Be confident but don't BS**. If the models don't work do not ad-lib, failures may tell everyone something important about the data.
3. **Question everything**. Never be afraid to ask "stupid questions"
4. **Don't game the system**. Decisions will be a balance of factors, don't try to get the answer you want by inflating your results.
5. **Listen to others**. There will always be people with deeper epidemiological insights.
6. **Beware the Cassandra effect, magnified by hindsight goggles**.
7. **It is not the modeller's role to set policy** – if you want to set policy go into politics.

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Principles | Proper scoring rules for evaluation

A scoring rule S(p,y) is proper if for any **true** probability distribution q:

$$E_{y\sim q}[S(q,y)] \leq E_{y\sim q}[S(p,y)]$$

the expected score $E_{y\sim q}$ is minimised when you report your q, rather than any other distribution p (lower score is better)

The forecaster can't "game the system". If q is true, they can't score better with any p.

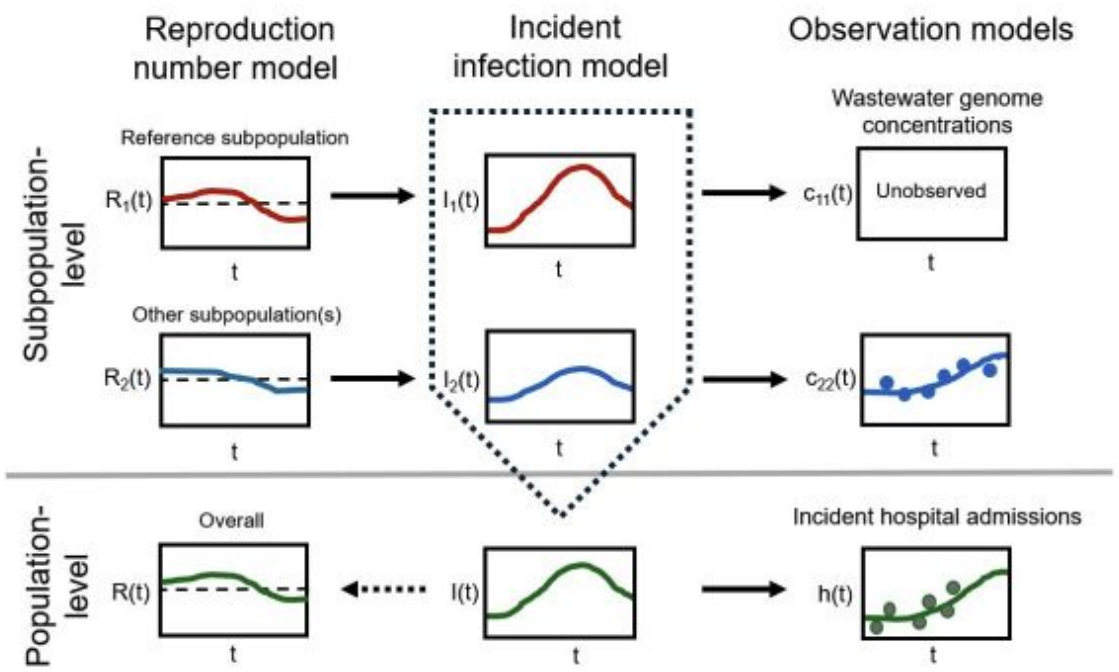
- Examples of proper scoring rules: Continuous ranked probability score (CRPS) for trajectories, Weighted Interval Score (WIS) for quantile-based forecasts, Brier scores for probability of categorical event occurrence
- Incentivises well-calibrated forecasts (maximise sharpness subject to calibration), allows for a fair comparison between forecasts

Gneiting, T. & Raftery A.E. Strictly Proper Scoring Rules, Prediction, and Estimation. 2007.

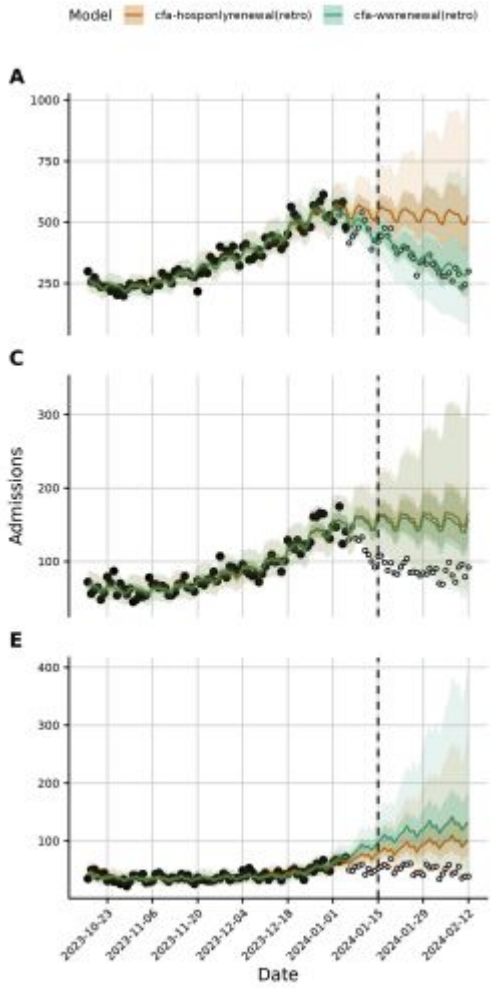
Practice | Model development in real-time operational setting

Anecdote on real-time model development at a public health agency

- Incorporating a new data stream
- Shown to be a leading indicator, thought was that it could improve forecast performance
- It's new, we don't have historical data to back-test
- We planned to use proper scoring rules and do a post-hoc retrospective evaluation



Johnson KE, et al. Bayesian generative modeling for heterogeneous wastewater data applied to COVID-19 forecasting. medRxiv. (2026).



We ultimately found no improvement from incorporating wastewater. Was our approach irresponsible?

What is our responsibility as modellers trying to inform policy decisions in real-time?

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- Are there general guidelines we should follow as modellers to ensure that we are being responsible in the models we developed and how the results from them may be used/interpreted?
  - Do these differ depending on our role e.g. in a public health agency or in academic setting?
- How do we enable rapid, real-time model development and adaptation while also acknowledging implicit assumptions about the robustness of modelling which may not be met for new models/model components?
  - Should we always compare to status quo/original model?
- Is post-hoc retrospective model evaluation sufficient, and what makes a model "good enough" for policy-making?
- How to evaluate model performance when none of the scenarios projected are what played out in reality?

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What is model error?

Appropriate choices of evaluation tools depend on your conception of model error.

Error = Reality minus Model = ?

Ethical basis for models. Trust, and value alignment with decision makers

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Speaker slides: Graham 1

IRRESPONSIBILITY

- Doing the modelling wrong
  - Reproducibility
  - Be careful crossing disciplines and adopting techniques
  - Defining the context as well as the content
- Assuming you are right
  - You probably aren't…
  - And neither are they…
- You are not making the decisions
  - See above
- Content and context
  - No model includes everything

Speaker slides: Graham 2

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RESPONSIBILITY

- Conduct the research with vigour, urgency and confidence
  - Leave no stone unturned
  - Excellence should be standard
  - Do the research like other people's' lives depend on it
- Present the work with candour and humility
  - TMWIW
  - You haven't thought of everything - inequalities?
  - Present it early and often - **encourage others to address the same questions**
  - Present it like you want the decision-makers to take it into account, but not swallow it blindly

- graham.medley@lshtm.ac.uk

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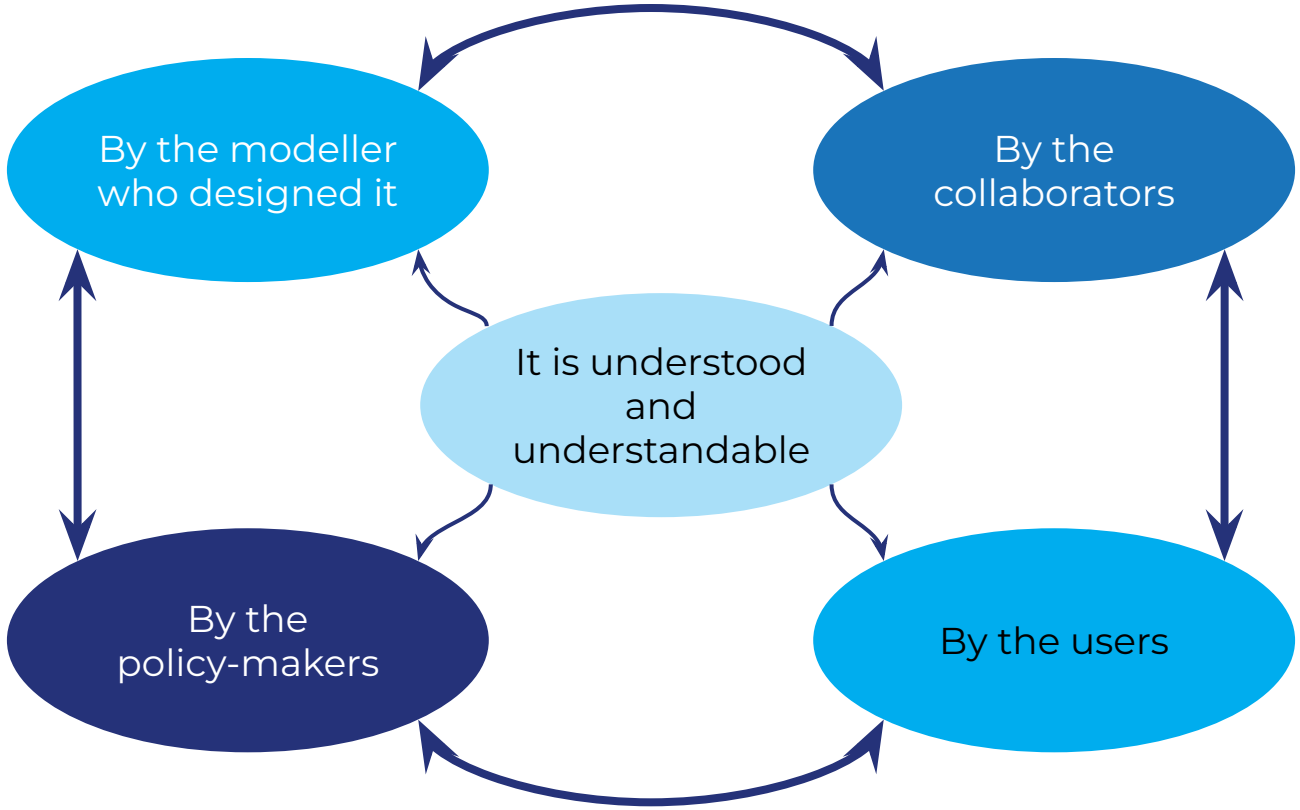
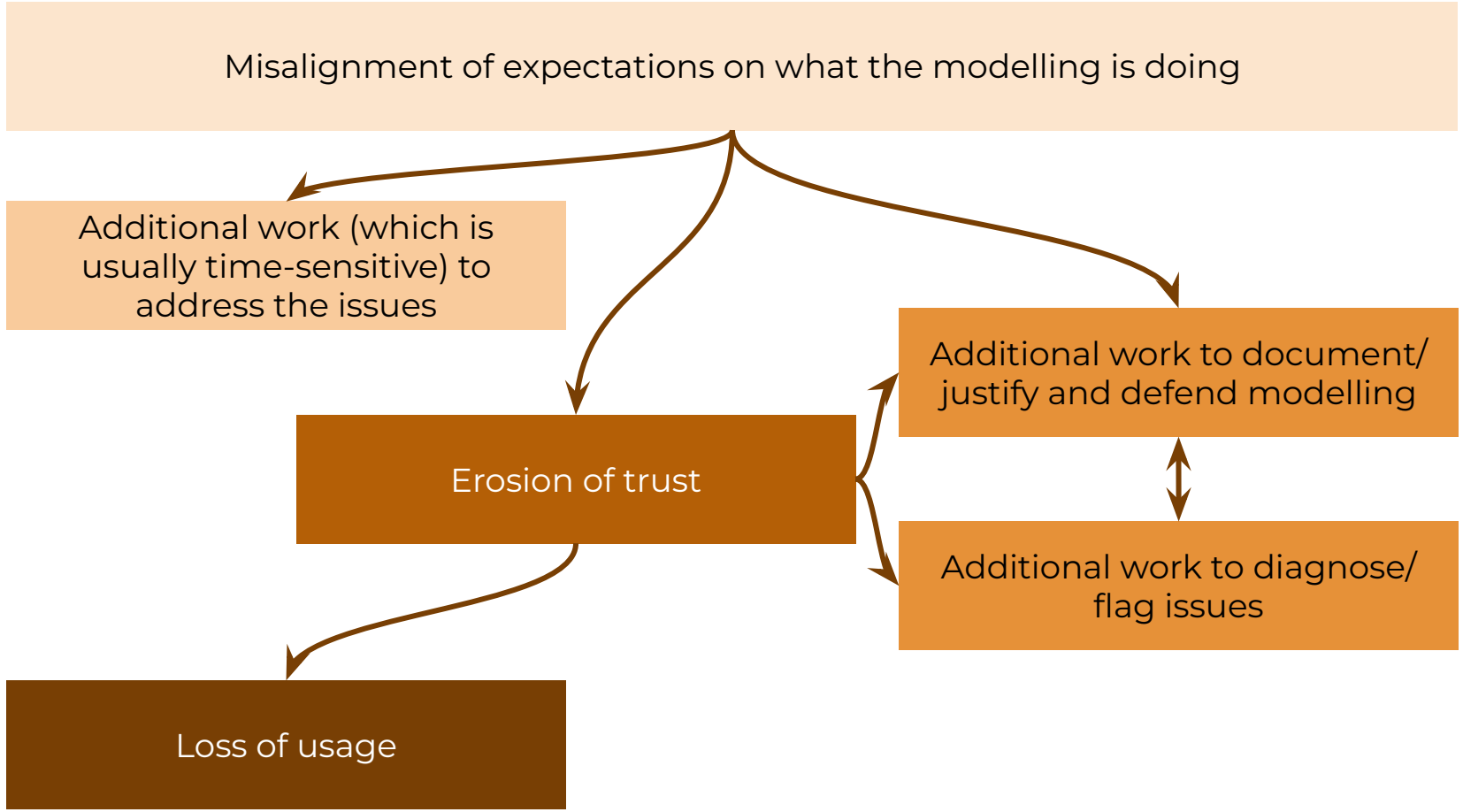
Speaker slides: Katy



Speaker slides: Katy



Speaker slides: Katy

Avoid complications

Diagnostics and reproducible pipelines are your friends and will save you time

Practise communication

Focus on your use-case

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Speaker slides: Juan

- Limited scope – a well defined question
- Timely response is essential
- Data rich environment
- A process involving several actors: data scientists, economists, field epidemiologists wet lab scientists
- Be ready to get your modelling output dismissed!
- Academic output is less of a priority

Speaker slides: Juan

- The field changes quite fast and sometimes revisions of models are part of process: PCV (pneumo)
- Sometimes the best model is the one available… on time!
- There are processes in place to secure robustness: 1st and 2nd opinion models for JCVI
- Sometimes models go wrong

Speaker slides: Juan

- The right question matters more than the most complex model
- Collaboration is embedded in the methodology: disease experts, surveillance specialists, and modelers must co-own the problem
- Data availability defines model scope, not the other way around
- Robustness means a model is *fit for purpose*, not perfect

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Priorities & actions

16:00 Crowd sourcing: big priorities and smallest next steps
Rapid rating exercise

16:30 Close

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Edits

- Afternoon “training” - instead asked “what would you do to ensure irresponsible modelling in this scenario?” (groups + room feedback)
- Smallest next step - swapped 3x for 30 sec each, rated /5